

1. Background

Steel manufacturing plants operate highly complex, capital-intensive, and interdependent equipment systems. Any unplanned downtime in such environments can result in production loss, safety risks, operational inefficiencies, and increased maintenance costs.

In practice, maintenance engineers often rely on multiple fragmented information sources—such as equipment manuals, standard operating procedures (SOPs), historical maintenance logs, failure analysis reports, and sensor-based alerts—to identify equipment issues and determine appropriate corrective action. This process is typically manual, time-consuming, and dependent on expert judgement, which can delay response times and reduce consistency in maintenance decisions.

There is a clear need for an intelligent, context-aware maintenance support system that can consolidate diverse data sources, assist in diagnosis, improve root cause analysis, recommend prioritized maintenance actions, and enable predictive maintenance for critical industrial assets.

2. Objective

The objective of this problem statement is to design and develop an intelligent Maintenance Wizard for industrial equipment in steel manufacturing environments.

The proposed system should act as a decision-support platform for maintenance engineers by enabling:

- Faster and more accurate diagnosis of equipment issues
- Identification of probable root causes of failures
- Prediction of equipment degradation and remaining useful life
- Proactive detection of abnormalities and catastrophic failure risks
- Prioritization of maintenance actions based on operational and procurement constraints
- Generation of structured maintenance insights and reports

The solution should support both reactive troubleshooting and proactive maintenance planning, thereby improving plant reliability, reducing downtime, and enhancing maintenance efficiency.

3. Problem Description

Participants are required to build a system that can ingest maintenance-related inputs from multiple sources and provide explainable, actionable, and structured outputs for maintenance decision-making.

The system should be capable of analysing equipment conditions, identifying abnormalities, assessing risk levels, recommending immediate and long-term maintenance actions, and assisting users through natural language interaction.

The proposed solution should also support learning from historical data and engineer feedback, enabling continuous improvement in recommendations and predictions over time.

4. Expected Inputs

The system should accept one or more of the following input types:

4.1 Operational and Failure Inputs

- Equipment delay logs
- Fault/error messages generated by equipment or control systems
- Failure analysis reports
- Incident records and historical breakdown summaries

4.2 Condition Monitoring Inputs

- Sensor data summaries
- Abnormality or anomaly alerts
- Process condition indicators relevant to equipment health

4.3 Knowledge and Documentation Inputs

- Equipment manuals
- Maintenance SOPs
- Historical maintenance records
- Spare parts information, including availability and procurement lead time

4.4 User Interaction Inputs

- Natural language queries from maintenance engineers
- Scenario-based learning or troubleshooting prompts
- Follow-up queries in a multi-turn conversational format

5. Expected Outputs

The system should generate clear, structured, and explainable outputs, including but not limited to the following:

5.1 Diagnostic and Predictive Outputs

- Probable fault diagnosis
- Root cause analysis of failures or abnormal conditions
- Prediction of remaining lifecycle / remaining useful life (RUL) of equipment or systems
- Early warning of potential catastrophic failures in critical equipment
- Detection of process-related defects contributing to equipment issues

5.2 Risk and Priority Outputs

- Risk level classification (e.g., low, medium, high, critical)
- Urgency assessment for intervention
- Bottleneck prioritization at the plant level
- Prioritization based on:
 - process criticality
 - delay severity
 - spares availability
 - procurement lead time

5.3 Maintenance Recommendation Outputs

- Step-by-step maintenance or repair recommendations
- Immediate action points
- Optimized maintenance plan
- Long-term monitoring recommendations
- Spare procurement strategy, where applicable

5.4 Reporting Outputs

- Structured maintenance reports
- Abnormal alert reports
- Maintenance decision summaries for engineers and supervisors
- Equipment-specific digital maintenance log entries (optional enhancement)

6. Functional Requirements

The proposed system should meet the following functional requirements:

1. Contextual Reasoning Using LLMs or SLMs

The solution should integrate Large Language Models (LLMs) or Small Language Models (SLMs) for contextual understanding, reasoning, and maintenance query answering. Additional merit may be given to teams that create or fine-tune a domain-specific model for industrial maintenance use cases. Publicly available APIs may be used.

2. Knowledge Integration

- a. The system should be capable of integrating and reasoning over:
- b. Equipment manuals
- c. Maintenance SOPs
- d. Historical maintenance records
- e. Failure reports and operational logs

3. Natural Language Interaction

- a. The solution should support natural language queries from users and handle multi-turn conversations in a context-aware manner.

4. Explainable Recommendations

- a. Outputs generated by the system should be explainable and traceable to relevant input data, historical records, rules, or documentation wherever possible.

5. Abnormality Detection and Failure Prediction

- a. The system should include an architecture for:
- b. Dynamic abnormality detection
- c. Early warning generation
- d. Failure prediction for critical equipment

6. Feedback-Driven Improvement

- a. The system should incorporate a feedback loop through which user corrections, confirmations, or outcomes can be used to improve future recommendations.

7. Real-Time Alerting Capability

- a. The solution should be capable of generating real-time abnormal alert reports and user-specific notifications where relevant.

7. Optional Enhancements

Teams may additionally implement one or more of the following features to enhance the usability and practical value of the system:

- Conversational interface for maintenance engineers
- Visualization dashboard for equipment health, trends, and anomalies
- Integration with simulated IoT or equipment monitoring dashboards
- Dynamic knowledge base for each equipment or system
- Automatic digital logbook for maintenance activities and observations
- User-role-based alerts and recommendations

8. Expected Outcome

The final solution should function as an intelligent maintenance decision-support system that helps industrial operations teams:

- Reduce unplanned downtime
- Improve maintenance response time
- Increase diagnostic accuracy
- Shift from reactive to proactive maintenance practices
- Improve maintenance planning and spare management
- Enable faster and more informed troubleshooting in industrial environments

The system should demonstrate practical applicability in a steel plant.

9. Deliverables

Participants are expected to submit the following:

- **Detailed Source Code of Working Prototype**
 - Complete code required to run the system.
- **A clear document explaining:**
 - System architecture
 - Technology stack used
 - Data flow and system flow
 - Model design and reasoning pipeline
 - Alerting and prediction logic
 - Assumptions and limitations
 - Documentation describing how to install, configure, and run the system
 - Sample Input and Output Demonstration
- **A screen recording that showcases the features that have been built**

All the above deliverables may be consolidated and uploaded as a single ZIP file.